BOARD OF INTERMEDIATE EDUCATION, KARACHI

INTERMEDIATE (11TH CLASS) EXAMINATION, 2024

MATHEMATICS

Past Paper 2024

Retyped faithfully from the original board paper — same questions, no changes.

Section 'A' (Multiple Choice Questions)

Maximum Marks: 40

Note: The section header says (40) but the note below it says 20 part questions; the scan shows 20 MCQs (i–xx), all transcribed. (Options in the original are separated by star marks, shown here as A–D in printed order.)

Note: Time: 3 Hours — Maximum Marks: 100 (Science Pre-Engineering Group)

Q1. Choose the correct answer for each from the given options:

Qi. Multiplicative inverse of $(-3, 4)$ is:

(A) $(3/25, 4/25)$ (B) $(3/25, -4/25)$

(C) $(-3/25, -4/25)$ (D) $(1, 0)$

Qii. Real part of $(2+i)/i$ is:

(A) -1 (B) 1

(C) $1/2$ (D) 2

Qiii. A square matrix A is skew − symmetric if :

(A) $A^t = A$ (B) $(\bar{A})^t = A$

(C) $A^t = -A$ (D) $A^t = |A|$

Qiv. If $A = [-7]$, then $|A| =$:

(A) 4 (B) 3

(C) 7 (D) -7

Qv. The number of non-zero rows in echelon form of matrix is called:

(A) Order of matrix (B) Leading rows

(C) Leading columns (D) Rank of a matrix

Qvi. If three vectors $\bar{a}$, $\bar{b}$ and $\bar{c}$ are coplanar, then $[\bar{a}\ \bar{b}\ \bar{c}]$ = :

(A) 0 (B) $\bar{a}$

(C) $\bar{b}$ (D) $\bar{c}$

Qvii. $\hat{k} \times \hat{j}$ = :

(A) $\hat{i}$ (B) $-\hat{i}$

(C) $\hat{j}$ (D) $-\hat{j}$

Qviii. No term in G.P is:

(A) 1 (B) -1

(C) 0 (D) i

Qix. If a = 3, d = 7 and $a_n = 59$ then n is equal to :

(A) 2 (B) 4

(C) 9 (D) 11

Qx. Σn is equal to:

(A) $n(n+1)(2n+1)/6$ (B) $n(n+1)/2$

(C) $[(n(n+1))/2]^2$ (D) $n(n+1)/4$

Qxi. If n = r, then ${}^nP_r$ = :

(A) $(n - r)!$ (B) 1

(C) 0 (D) n!

Qxii. $1 - x + x^2 - x^3 + \ldots\ldots =$:

(A) $(1 - x)^{-1}$ (B) $(1 + x)^{-1}$

(C) $(1 + x)^{-2}$ (D) $(1 - x)^{-2}$

Qxiii. One to one function is called:

(A) Inverse function (B) Surjective function

(C) Bijective function (D) Injective function

Qxiv. A function $f(x) = |x| - x^2$ is a/an:

(A) odd (B) Even

(C) Linear (D) Circular

Qxv. $(\sin(\theta/2) - \cos(\theta/2))^2$ = :

(A) $1 - \cos\theta$ (B) $1 + \cos\theta$

(C) $1 + \sin\theta$ (D) $1 - \sin\theta$

Qxvi. If f(x) = 2x + 5, then find f(2):

(A) 12 (B) 11

(C) 10 (D) 9

Qxvii. The region of solution in L.P problem is called:

(A) Infeasibie region (B) Unbounded region

(C) Infinite region (D) Feasible region

Qxviii. In any triangle ABC, Area of triangle is:

(A) 1/2 absinγ (B) absinγ

(C) 1/2 absinα (D) 1/2 absinβ

Qxix. The period of sin(x/2) =:

(A) π (B) 2π

(C) 3π (D) 4π

Qxx. $\sec[\sin^{-1}(-1/2)]$:

(A) 1/2 (B) √3/2

(C) 2/√3 (D) −2/√3

Section 'B' (Short-Answer Questions)

Maximum Marks: 40

> Note: Attempt any TEN part questions in all. All questions carry equal marks. (i.e. 4 marks of each part).

Q2.

(i) Solve the following complex equation. (x, y)(2, 3) = (−4, 7)

(ii) If A = [[−2,1,0],[−1,4,3],[0,8,5]] then find $A^2 - 5A + 4I$

(iii) Using properties of determine prove that: |(b+c, a, a), (b, c+a, b), (c, c, a+b)| = 4abc

(iv) Calculate the projection of $\vec{b}$ along $\vec{a}$ when $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$ and $\vec{b} = 3\hat{i} - \hat{j} - 5\hat{k}$

(v) Find Arithmetic, Geometric and Harmonic means between 1 and 9. Also verify that $AH = G^2$

(vi) Find the sum of the series 3 + 6 + 21 + 96 + 471 + ... to n terms.

(vii) How many words can be formed by 3 vowels and 4 consonants out of 5 vowels and 7 consonants?

(vii-OR) How many arrangements can be formed using all the letters of the word 'MATHEMATICS'?

(viii) Using the principle of mathematical induction for every positive integer n, prove that: $1/(1.2) + 1/(2.3) + 1/(3.4) + \ldots + 1/(n(n+1)) = n/(n+1)$

(ix) Find the middle term in the expansion of $(a/b - b/a)^{18}$ OR find the term independent of 'x' in $(2x + 1/x^2)^9$

(x) Let f: R → R be the function defined by $f(x) = \sqrt{(x+7)} : x \geq -7$, find the value of $f^{-1}(x)$

(xi) If $A + B + C = 180°$ then prove that $\cos((B+C)/2) = \sin(A/2)$

(xii) Find the smallest angle in ΔABC when a = 25cm, b = 18cm and c = 21cm

(xiii) Prove that $1/r^2 + 1/r_1^2 + 1/r_2^2 + 1/r_3^2 = (a^2+b^2+c^2)/\Delta^2$

(xiv) Soive the equation $\cos\theta - \theta = 0$ graphically for interval $-\pi/2 \leq \theta \leq \pi/2$.

(xiv-OR) Show that $2\tan^{-1}(2/3) = \cos^{-1}(5/13)$

(xv) Find the point of intersection of the following functions graphically: $f(x) = x + 4$ and $g(x) = x^2 - 6x + 10$

Section 'C' (Detailed Answer Questions)

Maximum Marks: 40

Note: Attempt any FIVE question from this section. All questions carry equal marks.

Q3.

Solve the non-homogenous system of linear equations using Gauss Elimination method: $-x + y + z = 0$, $x + 2y = 5$, $-3x + 2y - z = -2$

Q4.

If the 3rd term and the 7th term of a H.P are 2/9 and 2/25 respectively, find the sequence.

Q5.

In a three child family, find the probability of: (a) Three boys (b) Exactly two boys (c) At least one girl

Q6.

Use Binomial Theorem to show: $1 + 1/4 + (1.3)/(4.6) + (1.3.5)/(4.6.8) + \ldots = \sqrt[3]{4}$

Q7.

Find the volume of the tetrahedron whose vertices are A(2,1,8), B(3,2,9), C(2,1,4) and D(3,3,10)

(OR) Prove Hexagon Law of vector addition.

Q8.

Solve the Linear programming by graphical method when $x \geq 0$ and $y \geq 0$ Maximize $Z(x,y) = 30x + 36y$ subject to: $4x + 2y + 12$; $6x + 5y \leq 20$

Q9.

Prove fundamental law $\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$

(OR) Prove that: i) $\cos4\theta = 8\cos^4\theta - 8\cos^2\theta + 1$ (ii) $(\sin6\theta+\sin4\theta)/(\cos6\theta+\cos4\theta) = \tan5\theta$

Q10.

Find the general solution set of the trigonometric equation $\operatorname{cosec}\theta = \sqrt{3} + \cot\theta$ and also verify the solution.